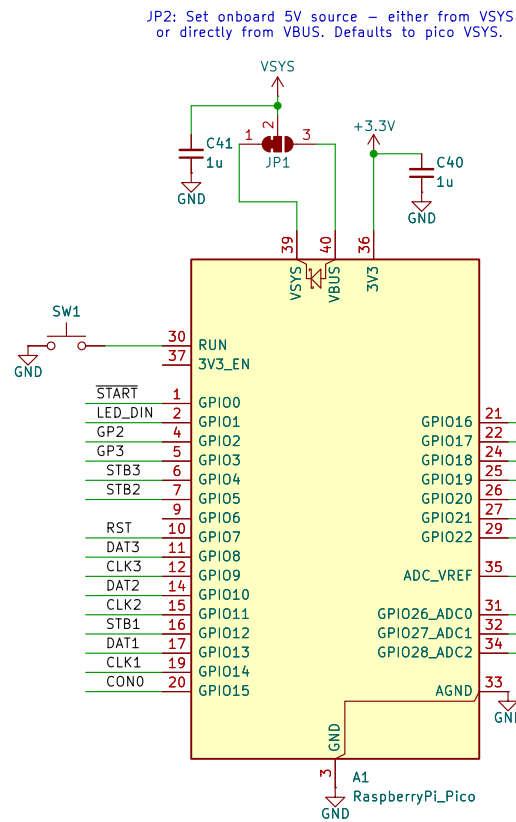
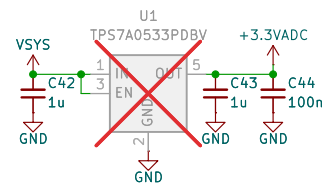


Raspberry Pico 2 module



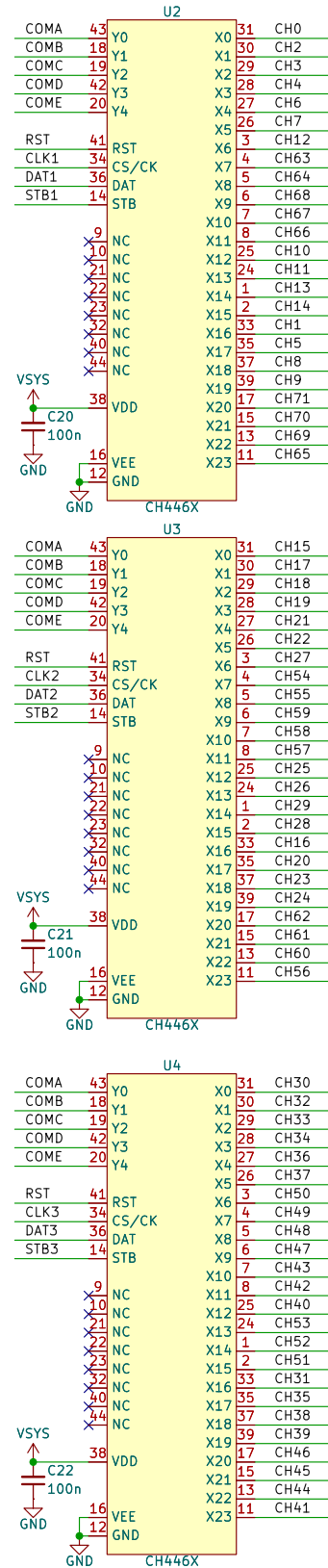
External ADC reference



REMOVE R7 on PICO2 when enabling external reference!!

Drive GPIO23 to improve ADC reference stability

3x MUX (72ch : 5)



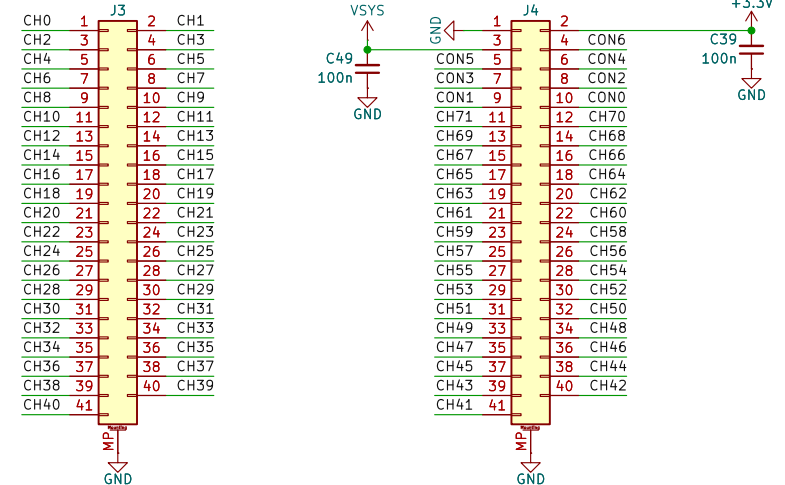
Adapter connector

2 x 41pin connectors:
82 total pins
72 connects to mux

HRS DF9-41P-1V(69)

Cheaper alternative:
XKB Connection X1016WF-41-46V01

CON6 is dedicated for adapter EEPROM
CON6 = EEPROM SIO

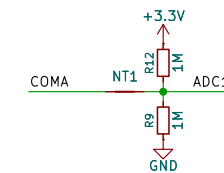


Current source & COMx mapping

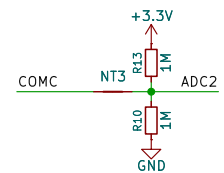
BuT – Bond under test

Current is injected into the DUT GND pad via COMD, forward-biasing the VSS-side diode. The Bond Under Test (BuT) is grounded via COMB, completing the current path. Neighbouring bonds (previous and next) are monitored to detect shorts to the BuT.

COMA – Previous bond (ADC1)



COMC – Next bond (ADC2)



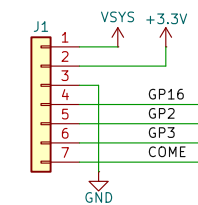
COMB – Bond-Under-Test (BUT)



COMD – Current source (ADC0)

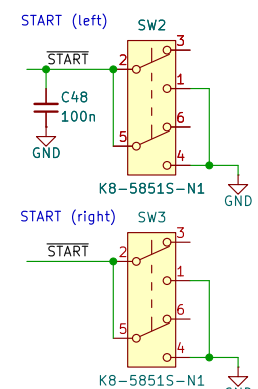
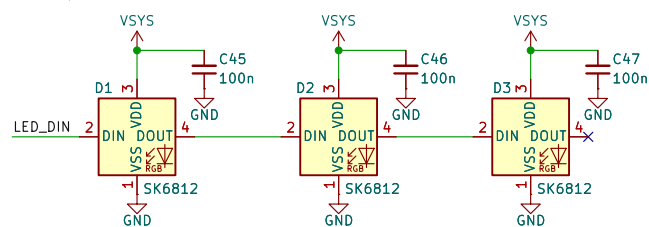


Extra pins breakout header



User interface

1x red LED
1x yellow LED
1x green LED
1x push button
1x aux input



Sheet: /

File: BondTest72.kicad_sch

Title:

Size: A3

Date:

Rev:

KiCad E.D.A. 10.0.1

Id: 1/1